DAY 5 | PRE-READ

The Web's Postal Service

A light read before Day 5. About 15 minutes, no code required.

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Why read this?

Yesterday you saw that clients talk to servers through APIs. Today you learn the language they speak, called HTTP, the rules that make an API pleasant to use, called REST, and you build your first API with FastAPI. Today also includes Basics assessment 1.

1. HTTP is like sending a letter

Letter	HTTP request	Example
Address on the envelope	URL	shop.com/products/42
What you want done	Method	Please send me (GET), please add (POST)
Notes on the envelope	Headers	Language, who you are
The letter inside	Body	Details of a new order
The reply letter	Response	Status code plus the information

2. Five common requests

- **GET:** show me something. Like looking at a menu.
- **POST:** create something new. Like placing an order.
- **PUT:** replace something completely. Like rewriting your address.
- **PATCH:** change one detail. Like fixing a typo in your address.
- **DELETE:** remove something. Like cancelling an order.

3. The reply always starts with a number

Every response comes with a three digit status code. You already know one: 404 Not Found.

Starts with	Meaning	Everyday version
2	Success	Your parcel was delivered.
4	Your mistake	Wrong address on the parcel.
5	Their mistake	The post office lost it.

4. Forms with rules

A good booking form refuses "abc" as the number of guests. In FastAPI, a tool called **Pydantic** lets you write these rules once. Bad data is turned away at the door, with a clear message explaining why.

Key idea: describe the shape of your data once, and let the tool check every request for you.

5. A helper that writes code NEW (v4)

Today you ask **OpenAI Codex** to write one endpoint for you. Think of it as a fast junior colleague: helpful, but you check its work. You read every line, run the tests, and only keep code you can explain.

6. About today's assessment

Today includes a 1.5 hour assessment on Python, OOP, MongoDB and FastAPI basics. Revise your post-reads and workbooks, and re-run the notebooks. Focus on understanding, not memorising.

7. Think about these before class

- When you search on a shopping site, which request type do you think your browser sends?
- What status code might you get if you try to open someone else's private account page?
- If a helper wrote code for you, how would you check it is correct?

Ganbatte kudasai (頑張ってください): please do your best, for the assessment